Problem 1a: Assuming that all matrices are $n \times n$ and invertible, solve for D:

$$ABC^TDBA^TC = AB^T.$$

Solution

Since A, B, and C are invertible, their transposes A^T, B^T, and C^T are also invertible. Moreover, the product of invertible matrices is also invertible. Therefore, the matrix ABC^T and BA^TC are invertible.
We solve for D by multiplying both sides appropriately by inverses:

$$\begin{aligned} ABC^TDBA^TC &= AB^T \\ (ABC^T)^{-1}ABC^TDBA^TC(BA^TC)^{-1} &= (ABC^T)^{-1}AB^T(BA^TC)^{-1} \\ D &= C^{-T}B^{-1}A^{-1}AB^TC^{-1}A^{-T}B^{-1} \\ \Rightarrow D &= C^{-T}B^{-1}B^TC^{-1}A^{-T}B^{-1} \end{aligned}$$

Problem 1b: For what values of k is the following system consistent?

$$kx + y = 1, \quad x + ky = 1$$

Solution

We write the system as a matrix equation:

$$\begin{bmatrix} k & 1 \\ 1 & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The system is consistent if the coefficient matrix is invertible or if the equations are consistent when it is not.
Compute the determinant of the matrix:

$$\det = k^2 - 1$$

Case 1: If $k^2 - 1 \neq 0 \Rightarrow k \neq \pm 1$, the matrix is invertible, so the system has a unique solution $\Rightarrow$ consistent.
Case 2: If $k = 1$, the system becomes:

$$x + y = 1, \quad x + y = 1 \quad \Rightarrow \text{Infinitely many solutions}$$

Case 3: If $k = -1$, the system becomes:

$$-x + y = 1, \quad x - y = 1 \Rightarrow \text{Add: } 0 = 2 \Rightarrow \text{Inconsistent}$$

Conclusion: The system is consistent for all $k \in \mathbb{R}$ except $k = -1$.

Problem 2: State and prove Cramer's rule for a 2×2 system. Then explain under what conditions the rule fails.

Cramer's Rule for a 2×2 System

Statement: For the system

$$a_1x + b_1y = c_1$$
$$a_2x + b_2y = c_2$$

let

$$A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}, \quad A_x = \begin{bmatrix} c_1 & b_1 \\ c_2 & b_2 \end{bmatrix}, \quad A_y = \begin{bmatrix} a_1 & c_1 \\ a_2 & c_2 \end{bmatrix}$$

Then, if $\det(A) \neq 0$, the unique solution is

$$x = \frac{\det(A_x)}{\det(A)}, \quad y = \frac{\det(A_y)}{\det(A)}.$$

Proof via Elementary Row Operations: Write the augmented matrix:

$$\left[\begin{array}{cc|c} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{array}\right]$$

Eliminate x from the second row:

$$R_2 \leftarrow R_2 - \frac{a_2}{a_1}R_1 \quad (\text{assuming } a_1 \neq 0)$$

$$\Rightarrow \left[\begin{array}{cc|c} a_1 & b_1 & c_1 \\ 0 & b_2 - \frac{a_2}{a_1}b_1 & c_2 - \frac{a_2}{a_1}c_1 \end{array}\right]$$

Let $D = a_1b_2 - a_2b_1 = \det(A)$.
Now solve for y from the second row:

$$y = \frac{a_1c_2 - a_2c_1}{D} = \frac{\det(A_y)}{\det(A)}$$

Substitute into the first row to find x:

$$x = \frac{c_1 - b_1y}{a_1} = \frac{c_1b_2 - c_2b_1}{D} = \frac{\det(A_x)}{\det(A)}$$

Failure Condition: Cramer's Rule fails if $\det(A) = 0$. In that case, the coefficient matrix is not invertible.

Problem 3: Let V be a finite-dimensional vector space, and let $S = \{v_1, v_2, \ldots, v_n\}$ be any basis. Prove that:

(a) Any set of more than n vectors in V is linearly dependent.

(b) Any set of fewer than n vectors in V does not span V.

Solution (a)

Let $S' = \{w_1, w_2, \ldots, w_m\} \subset V$, where $m > n$. Since $S = \{v_1, v_2, \ldots, v_n\}$ is a basis, each w_i can be expressed as a linear combination of the basis vectors:

$$\begin{aligned} w_1 &= a_{11}v_1 + a_{21}v_2 + \cdots + a_{n1}v_n \\ w_2 &= a_{12}v_1 + a_{22}v_2 + \cdots + a_{n2}v_n \\ &\vdots \\ w_m &= a_{1m}v_1 + a_{2m}v_2 + \cdots + a_{nm}v_n \end{aligned} \tag{1}$$

To show that S' is linearly dependent, we must find scalars $k_1, \ldots, k_m$, not all zero, such that:

$$k_1w_1 + k_2w_2 + \cdots + k_mw_m = 0 \tag{2}$$

Substituting from (1), we obtain:

$$\begin{aligned} &(k_1a_{11} + k_2a_{12} + \cdots + k_ma_{1m})\, v_1 \\ &+ (k_1a_{21} + k_2a_{22} + \cdots + k_ma_{2m})\, v_2 \\ &\vdots \\ &+ (k_1a_{n1} + k_2a_{n2} + \cdots + k_ma_{nm})\, v_n = 0 \end{aligned}$$

Since $\{v_1, \ldots, v_n\}$ is linearly independent, each coefficient must be zero:

$$\begin{aligned} a_{11}k_1 + a_{12}k_2 + \cdots + a_{1m}k_m &= 0 \\ a_{21}k_1 + a_{22}k_2 + \cdots + a_{2m}k_m &= 0 \\ &\vdots \\ a_{n1}k_1 + a_{n2}k_2 + \cdots + a_{nm}k_m &= 0 \end{aligned} \tag{3}$$

This is a homogeneous system of n equations in $m > n$ unknowns. Implies that a nontrivial solution exists. Hence, S' is linearly dependent.

Problem 3: Let V be a finite-dimensional vector space, and let $S = \{v_1, v_2, \ldots, v_n\}$ be any basis. Prove that:

(a) Any set of more than n vectors in V is linearly dependent.

(b) Any set of fewer than n vectors in V does not span V.

Solution (b)

Let $S' = \{w_1, w_2, \ldots, w_m\} \subset V$ with $m < n$. Suppose S' spans V. Then every $v_i \in S = \{v_1, \ldots, v_n\}$ can be written as a linear combination of $w_1, \ldots, w_m$:

$$\begin{aligned} v_1 &= a_{11}w_1 + a_{21}w_2 + \cdots + a_{m1}w_m \\ v_2 &= a_{12}w_1 + a_{22}w_2 + \cdots + a_{m2}w_m \\ &\vdots \\ v_n &= a_{1n}w_1 + a_{2n}w_2 + \cdots + a_{mn}w_m \end{aligned} \tag{4}$$

Now consider the linear combination:

$$k_1v_1 + k_2v_2 + \cdots + k_nv_n = 0 \tag{5}$$

Substitute from (4):

$$\sum_{j=1}^{n} k_j v_j = \sum_{j=1}^{n} k_j \left(\sum_{i=1}^{m} a_{ij} w_i \right) = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} k_j \right) w_i = 0$$

Same arguments follows from (a) to get:

$$\begin{aligned} a_{11}k_1 + a_{12}k_2 + \cdots + a_{1n}k_n &= 0 \\ a_{21}k_1 + a_{22}k_2 + \cdots + a_{2n}k_n &= 0 \\ &\vdots \\ a_{m1}k_1 + a_{m2}k_2 + \cdots + a_{mn}k_n &= 0 \end{aligned}$$

This system has $m < n$ equations in n unknowns. A nontrivial solution exists, contradicting the assumption that $\{v_1, \ldots, v_n\}$ is linearly independent.
Therefore, S' cannot span V.

Problem 4a: If $\mathbf{u}$ and $\mathbf{v}$ are $n \times 1$ matrices and A is an $n \times n$ matrix, then show that

$$(\mathbf{v}^T A^T A\mathbf{u})^2 \leq (\mathbf{u}^T A^T A\mathbf{u})(\mathbf{v}^T A^T A\mathbf{v}).$$

Solution

Define new vectors $\mathbf{x} = A\mathbf{u}$ and $\mathbf{y} = A\mathbf{v}$. Then the left-hand side becomes:

$$(\mathbf{v}^T A^T A\mathbf{u}) = (A\mathbf{v})^T(A\mathbf{u}) = \mathbf{y}^T\mathbf{x} = \mathbf{x} \cdot \mathbf{y},$$

so:

$$(\mathbf{v}^T A^T A\mathbf{u})^2 = (\mathbf{x}^T\mathbf{y})^2 = |\mathbf{x} \cdot \mathbf{y}|^2.$$

Now apply the standard Cauchy–Schwarz inequality in $\mathbb{R}^n$:

$$|(\mathbf{x} \cdot \mathbf{y}|^2 \leq \|\mathbf{x}\|^2\|\mathbf{y}\|^2$$
$$(\mathbf{x}^T\mathbf{y})^2 \leq (\mathbf{x}^T\mathbf{x})(\mathbf{y}^T\mathbf{y}).$$

Substitute back:

$$\mathbf{x}^T\mathbf{x} = (A\mathbf{u})^T(A\mathbf{u}) = \mathbf{u}^T A^T A\mathbf{u}, \quad \mathbf{y}^T\mathbf{y} = (A\mathbf{v})^T(A\mathbf{v}) = \mathbf{v}^T A^T A\mathbf{v}.$$

Hence,

$$(\mathbf{v}^T A^T A\mathbf{u})^2 \leq (\mathbf{u}^T A^T A\mathbf{u})(\mathbf{v}^T A^T A\mathbf{v}).$$

Problem 4b: Prove that a system of linear equations $AX = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is in the column space of A.

Solution

Let $A = [\mathbf{c}_1 \ \mathbf{c}_2 \ \cdots \ \mathbf{c}_n]$, where $\mathbf{c}_i$ are the column vectors of A, and let $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$.

Then the matrix product:

$$AX = x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + \cdots + x_n\mathbf{c}_n.$$

So the equation $AX = \mathbf{b}$ becomes:

$$x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + \cdots + x_n\mathbf{c}_n = \mathbf{b}.$$

This means that $\mathbf{b}$ is a linear combination of the columns of A, i.e., $\mathbf{b} \in \text{Col}(A)$.
Conclusion: The system $AX = \mathbf{b}$ has a solution (is consistent) if and only if $\mathbf{b}$ is in the column space of A.

Problem 5: Let T_A be the linear transformation defined by the matrix below. Calculate $\ker(T_A)$.

$$A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$$

Solution

To find $\ker(T_A)$, solve $A\mathbf{x} = 0$, where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$.

$$\left[\begin{array}{cccc|c} 1 & 4 & 5 & 2 & 0 \\ 2 & 1 & 3 & 0 & 0 \\ -1 & 3 & 2 & 2 & 0 \end{array}\right] \quad R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 + R_1$$

$$\left[\begin{array}{cccc|c} 1 & 4 & 5 & 2 & 0 \\ 0 & -7 & -7 & -4 & 0 \\ 0 & 7 & 7 & 4 & 0 \end{array}\right] \quad R_3 \leftarrow R_3 + R_2$$

$$\left[\begin{array}{cccc|c} 1 & 4 & 5 & 2 & 0 \\ 0 & -7 & -7 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \quad R_2 \leftarrow -\tfrac{1}{7}R_2$$

$$\left[\begin{array}{cccc|c} 1 & 4 & 5 & 2 & 0 \\ 0 & 1 & 1 & \frac{4}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \quad R_1 \leftarrow R_1 - 4R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & -\frac{2}{7} & 0 \\ 0 & 1 & 1 & \frac{4}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

From the final matrix:

$$\begin{aligned} x_1 &= -x_3 + \frac{2}{7}x_4 \\ x_2 &= -x_3 - \frac{4}{7}x_4 \end{aligned} \quad \Rightarrow \quad \mathbf{x} = x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} \frac{2}{7} \\ -\frac{4}{7} \\ 0 \\ 1 \end{bmatrix}$$

$$\ker(T_A) = \text{span}\left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{2}{7} \\ -\frac{4}{7} \\ 0 \\ 1 \end{bmatrix} \right\}$$

Problem 6: If A and B are orthogonal matrices of the same order, then show that:

1. AB is orthogonal,
2. A^T is orthogonal,
3. A^{-1} is orthogonal.

Solution

(i) Since A and B are orthogonal, we have:

$$A^TA = I \quad \text{and} \quad B^TB = I$$

Consider $(AB)^T(AB)$:

$$(AB)^T(AB) = B^TA^TAB = B^TIB = B^TB = I$$

Hence, AB is orthogonal.
(ii) Since A is orthogonal, we know:

$$A^TA = I \Rightarrow AA^T = I$$

This implies:

$$(A^T)^TA^T = AA^T = I$$

So, A^T is orthogonal.
(iii) For an orthogonal matrix A, we know:

$$A^T = A^{-1}$$

But from part (ii), A^T is orthogonal. Therefore, A^{-1} is also orthogonal.

Problem 7a: Show that if A is an $n \times n$ orthogonal matrix, and $\langle A\mathbf{u}, A\mathbf{v}\rangle = A\mathbf{u} \cdot A\mathbf{v}$, then $\langle A\mathbf{u}, A\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle$.

Solution

Let $A \in \mathbb{R}^{n\times n}$ be an orthogonal matrix. Then by definition:

$$A^TA = I.$$

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. We compute:

$$\langle A\mathbf{u}, A\mathbf{v}\rangle = (A\mathbf{u})^T(A\mathbf{v}) = \mathbf{u}^TA^TA\mathbf{v}.$$

Since $A^TA = I$, we get:

$$\mathbf{u}^TA^TA\mathbf{v} = \mathbf{u}^T\mathbf{v} = \langle \mathbf{u}, \mathbf{v}\rangle.$$

Conclusion:

$$\langle A\mathbf{u}, A\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle.$$

That is, orthogonal matrices preserve inner products.

Problem 7b: Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation with $T(\mathbf{e}_1) = (1, 1)$, $T(\mathbf{e}_2) = (3, 0)$, and $T(\mathbf{e}_3) = (4, -7)$. Find the matrix of T.

Solution

The standard basis of $\mathbb{R}^3$ is $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$. For any $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3 \in \mathbb{R}^3$, linearity of T gives:

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) + x_3T(\mathbf{e}_3).$$

Thus, the matrix of T has columns $T(\mathbf{e}_1), T(\mathbf{e}_2), T(\mathbf{e}_3)$, since:

$$T(\mathbf{x}) = [T] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Given:

$$T(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_3) = \begin{bmatrix} 4 \\ -7 \end{bmatrix},$$

we get:

$$[T] = \begin{bmatrix} 1 & 3 & 4 \\ 1 & 0 & -7 \end{bmatrix}.$$

Problem 8: If A is an $n \times n$ matrix, and A has n linearly independent eigenvectors, then prove that A is diagonalizable.

Solution

Let $A \in \mathbb{R}^{n \times n}$ have n linearly independent eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$, with corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$, so

$$A\mathbf{v}_i = \lambda_i\mathbf{v}_i \quad \text{for } i = 1, \ldots, n.$$

Define $P = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n]$. Since the $\mathbf{v}_i$'s are linearly independent, P is invertible. Then:

$$AP = A[\mathbf{v}_1 \ \cdots \ \mathbf{v}_n] = [A\mathbf{v}_1 \ \cdots \ A\mathbf{v}_n] = [\lambda_1\mathbf{v}_1 \ \cdots \ \lambda_n\mathbf{v}_n] = PD,$$

where

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

Thus,

$$A = PDP^{-1},$$

so A is similar to a diagonal matrix and hence diagonalizable.

Problem 9a: Show that the set of all polynomials $p(x)$ such that $p(-x) = p(x)$ is a subspace of the vector space $\mathbb{P}_n$ of all polynomials of degree at most n.

Solution

Let $W = \{p(x) \in \mathbb{P}_n : p(-x) = p(x)\}$. We will show that W is a subspace of $\mathbb{P}_n$ by verifying the three conditions:
1. Zero polynomial is in W: The zero polynomial satisfies $0(-x) = 0 = 0(x)$, so $0 \in W$.
2. Closed under addition: Let $p(x), q(x) \in W$, so $p(-x) = p(x)$ and $q(-x) = q(x)$. Then:

$$(p+q)(-x) = p(-x) + q(-x) = p(x) + q(x) = (p+q)(x),$$

so $p + q \in W$.
3. Closed under scalar multiplication: Let $p(x) \in W$, and $c \in \mathbb{R}$. Then:

$$(cp)(-x) = c \cdot p(-x) = c \cdot p(x) = (cp)(x),$$

so $cp \in W$.

Problem 9b: Show that the set of all polynomials $p(x) \in \mathbb{P}_n$ such that $p'(0) = 0$ is a subspace of $\mathbb{P}_n$. Find a basis for this subspace.

Solution

Let $W = \{p(x) \in \mathbb{P}_n : p'(0) = 0\}$. We will verify the subspace properties and then find a basis.
1. Zero polynomial: $p(x) = 0 \Rightarrow p'(x) = 0 \Rightarrow p'(0) = 0$, so $0 \in W$.
2. Closed under addition: Let $p(x), q(x) \in W$, so $p'(0) = 0$, $q'(0) = 0$. Then:

$$(p+q)'(0) = p'(0) + q'(0) = 0 + 0 = 0 \Rightarrow p + q \in W.$$

3. Closed under scalar multiplication: Let $c \in \mathbb{R}$, $p(x) \in W$. Then:

$$(cp)'(0) = c \cdot p'(0) = c \cdot 0 = 0 \Rightarrow cp \in W.$$

Thus, W is a subspace of $\mathbb{P}_n$.
Basis: Let the standard basis of $\mathbb{P}_n$ be $\{1, x, x^2, \ldots, x^n\}$. Then for any $p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, we have:

$$p'(x) = a_1 + 2a_2x + \cdots + na_nx^{n-1} \Rightarrow p'(0) = a_1.$$

So $p'(0) = 0 \Rightarrow a_1 = 0$. Hence, the general form of elements in W is:

$$p(x) = a_0 + a_2x^2 + a_3x^3 + \cdots + a_nx^n.$$

Thus, a basis for W is:

$$\{1, x^2, x^3, \ldots, x^n\}.$$

Problem 10a: Let u be a vector in an inner product space V, and let $\{v_1, v_2, \ldots, v_n\}$ be an orthonormal basis for V. Show that if α_i is the angle between u and v_i, then

$$\cos^2 \alpha_1 + \cos^2 \alpha_2 + \cdots + \cos^2 \alpha_n = 1.$$

Solution

Since $\{v_1, \ldots, v_n\}$ is an orthonormal basis, we can express $u \in V$ as:

$$u = \sum_{i=1}^{n} \langle u, v_i \rangle v_i.$$

Then the squared norm of u is:

$$\|u\|^2 = \langle u, u \rangle = \left\langle \sum_{i=1}^{n} \langle u, v_i \rangle v_i,\ \sum_{j=1}^{n} \langle u, v_j \rangle v_j \right\rangle.$$

Using linearity and orthonormality $\langle v_i, v_j \rangle = \delta_{ij}$, we get:

$$\|u\|^2 = \sum_{i=1}^{n} |\langle u, v_i \rangle|^2.$$

Now by definition, $\cos \alpha_i = \dfrac{\langle u, v_i \rangle}{\|u\|}$. Hence:

$$\cos^2 \alpha_i = \frac{|\langle u, v_i \rangle|^2}{\|u\|^2}.$$

So:

$$\sum_{i=1}^{n} \cos^2 \alpha_i = \sum_{i=1}^{n} \frac{|\langle u, v_i \rangle|^2}{\|u\|^2} = \frac{1}{\|u\|^2} \sum_{i=1}^{n} |\langle u, v_i \rangle|^2 = \frac{\|u\|^2}{\|u\|^2} = 1.$$

Conclusion:

$$\cos^2 \alpha_1 + \cos^2 \alpha_2 + \cdots + \cos^2 \alpha_n = 1.$$

Problem 10b: A square matrix A is called nilpotent if $A^n = 0$ for some positive integer n. What can you say about the eigenvalues of a nilpotent matrix?

Solution

Let $A \in \mathbb{R}^{n \times n}$ be a nilpotent matrix, so there exists some $k \in \mathbb{N}$ such that:

$$A^k = 0.$$

Let λ be an eigenvalue of A, and let $\mathbf{v} \neq 0$ be a corresponding eigenvector, so:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Applying A^k repeatedly:

$$A^2\mathbf{v} = \lambda^2\mathbf{v}, \quad A^3\mathbf{v} = \lambda^3\mathbf{v}, \quad \ldots, \quad A^k\mathbf{v} = \lambda^k\mathbf{v}.$$

But since $A^k = 0$, it follows that:

$$A^k\mathbf{v} = 0 \Rightarrow \lambda^k\mathbf{v} = 0.$$

Because $\mathbf{v} \neq 0$, this implies $\lambda^k = 0 \Rightarrow \lambda = 0$.
Conclusion: All eigenvalues of a nilpotent matrix are zero.